

Advanced Robotics Using AI And IoT

8-Week Professional Virtual Internship Project Presentation & Technical Summary

CANDIDATE CREDENTIALS

- Name: Roshan Kshetri
- Program: B.Tech (Computer Science & Engineering)
- Specialization: AIML
- Batch: 2026–2027
- Institution: IILM University, Greater Noida

INTERNSHIP PROGRAM DETAILS

- Training Partner: EduSkills Academy (ISO 9001:2015)
- Patronage: AICTE National Internship Portal
- Duration: 8 Weeks (April 2026 – June 2026)
- Offer Ref: C16Y2026M091396204
- AICTE Student ID: STU69e08d9e3f2e51776323998
- Certificate ID: 2026-DFEBBA08EF (Issued Sep 2026)

Candidate & Program Overview

ACADEMIC PROFILE

Student Name: Roshan Kshetri
Degree Program: Bachelor of Technology (B.Tech)
Core Branch: Computer Science and Engineering
Specialization Track: Cyber Security
Current Academic Batch: 2026–2027
Department / School: School of Computer Science and Engineering
University: IILM University, Greater Noida, Uttar Pradesh
Focus Areas: Cyber-Physical Systems, Embedded Security, Network Telemetry, Edge Computing

INTERNSHIP SPECIFICATIONS

Internship Track: Advanced Robotics Using AI And IoT
Credential Designation: Robotics Fundamentals - AI and IoT Integrated
Program Mode: 8-Week Virtual Internship (Simulation & Project Labs)
Tenure / Duration: April 2026 – June 2026
Offer Letter Ref: C16Y2026M091396204
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Certificate ID: 2026-DFEBBA08EF (Sep 10, 2026)
Evaluation Scheme: Weekly Milestone Assessments + Documentation + Final Comprehensive Examination

Organization Profile: EduSkills Academy & AICTE

Nation Building Through Skills — Strategic Technical Education & Industrial Enablement

AICTE Patronage & Alignment

- Direct collaboration with Ministry of Education & AICTE.
- Integrated with the AICTE National Internship Portal.
- Enhances nationwide engineering employability.
- Certified under ISO 9001:2015 & ISO 20000-1:2018.

Industry Partnerships

- Strategic global alliances with AWS Academy, Red Hat, Cisco, VMware, Microchip.
- Curricula aligned with enterprise production standards.
- Hands-on virtual labs, simulation tools, and real-world architectures.

Core Domain Focus

- Embedded microcontrollers (Arduino, RP2040, ESP32).
- Sensor fusion & closed-loop kinematic motor control.
- Edge machine learning & computer vision pipelines.
- Scalable IoT telemetry & MQTT cloud communication.

Problem Statement & Industry Context

THE INDUSTRY CHALLENGE

- **Siloed Hardware & Software:** Academic robotics frequently isolates electronic circuit design from high-level computational cognition.
- **Centralized Cloud Latency:** Offloading robotic perception to remote cloud servers causes unacceptable communication latency in time-critical navigation.
- **Telemetry Blind Spots:** Industrial mobile robots often operate in isolated silos without standardized IoT telemetry streaming to central dashboards.
- **Cyber-Physical Risks:** Networked autonomous nodes are vulnerable to actuator hijacking, sensor data spoofing, and insecure firmware tampering.

APPLIED SOLUTION & SCOPE

- **Edge Intelligence:** Deploy lightweight computer vision & quantized ML perception directly onto embedded edge computing modules.
- **Closed-Loop Real-Time Control:** Microcontroller-driven PWM actuation with PID feedback loops ensuring sub-millisecond motor responsiveness.
- **Lightweight IoT Telemetry:** Asynchronous MQTT publish-subscribe messaging over Wi-Fi streaming live sensor data to supervisory dashboards.
- **Hardened Cyber Security:** Integration of TLS encryption, token-based command authentication, watchdog fail-safes, and emergency shutdown interlocks.

Core Project Objectives

1. Hardware & Circuit Interfacing

- Master electronic fundamentals: Ohm's Law, voltage dividers, filtering.
- Interface ultrasonic (HC-SR04), IR arrays, and 6-DOF IMU sensors.
- Design isolated dual-rail power supplies to eliminate motor back-EMF noise.

2. Actuation & Kinematic Control

- Implement H-bridge motor drivers (L298N) with PWM speed regulation.
- Formulate differential-drive forward & inverse kinematic models.
- Tune closed-loop PID control loops for trajectory tracking and obstacle avoidance.

3. Edge AI Perception & Vision

- Integrate OpenCV image processing pipelines for real-time visual tracking.
- Deploy machine learning classifiers for landmark & obstacle categorization.
- Execute multi-sensor fusion combining ultrasonic ranging with optical frames.

4. IoT Telemetry & Cyber Security

- Connect embedded ESP32/RP2040 microcontrollers to MQTT cloud brokers.
- Stream real-time diagnostic telemetry packets (battery, motor RPM, distance).
- Implement TLS socket encryption, command verification, and fail-safe watchdogs.

Technologies & Robotics Toolchain

Embedded & Hardware

- Arduino Uno / Nano
- Raspberry Pi Pico
- ESP32 SoC (Wi-Fi/BLE)
- L298N Dual H-Bridge
- DC Gear & Servo Motors
- HC-SR04 & MPU6050 IMU

Software & Scripting

- Python 3.x / MicroPython
- C++ / Embedded C
- Linux (Ubuntu CLI)
- Bash Scripting
- VS Code & PlatformIO
- Git & GitHub Versioning

AI, CV & Simulation

- OpenCV Computer Vision
- Scikit-Learn & NumPy
- TensorFlow Lite Micro
- Webots Robot Simulator
- Gazebo Simulation
- Wokwi Virtual Lab

IoT & Security

- MQTT Broker (Mosquitto)
- HTTP / REST APIs
- WebSockets Telemetry
- TLS 1.3 / SSL Encryption
- Hardware Watchdog Timers
- Fail-Safe Safe States

Production Robotics Architecture (6-Layer Model)

Layer 6: Supervisory Cloud & UI Layer

Web Telemetry Dashboard, Real-time Gauges, Anomaly Heatmaps, Remote Emergency Overrides

Layer 5: IoT & Network Communication Layer

Secure MQTT Broker, TLS-Encrypted Sockets, Wi-Fi 802.11 b/g/n, Bi-directional Command Dispatch

Layer 4: Edge AI & Perception Layer

OpenCV Color/Edge Segmentation, Lightweight ML Target Identification, Multi-Sensor Fusion

Layer 3: Kinematics & Motion Control Layer

Differential-Drive Kinematics (v, w), Closed-Loop PID RPM Regulation, Dynamic Collision Avoidance

Layer 2: Embedded Firmware & Real-Time Layer

100 Hz Control Loop, Interrupt-Driven ADC Sampling, Hardware PWM Modulation, Watchdog Timers

Layer 1: Physical Sensing & Actuation Layer

Chassis, DC Gear Motors, SG90 Servos, HC-SR04 Sonar, IR Arrays, MPU6050 IMU, Isolated Power Rails

Curriculum Timeline & Milestones (Weeks 1 to 4)

WEEK 1

Robotics Concepts & Basic Electronics

- Overview of autonomous mobile robotics.
- Ohm's Law, power calculations, circuit schematics.
- Voltage division, pull-up/down resistors.
- Breadboard circuit prototyping & multimeter testing.

WEEK 2

Microcontrollers & Embedded Sensors

- Microcontroller architectures (ATmega328P / RP2040).
- Digital GPIOs, Analog ADC sampling, Interrupts.
- Ultrasonic ranging (HC-SR04) calibration.
- Infrared line and obstacle sensor arrays.

WEEK 3

Actuators & Robotics Programming

- Operating mechanics of DC motors & servos.
- Dual H-Bridge motor drivers (L298N).
- Hardware PWM speed & directional control.
- Modular C++ & Python robot control scripting.

WEEK 4

Kinematics & Foundations of AI

- Differential-drive robot kinematic models.
- Forward & inverse kinematics, wheel odometry.
- Closed-loop PID speed feedback regulation.
- State machines & reactive artificial intelligence.

Curriculum Timeline & Milestones (Weeks 5 to 8)

WEEK 5

Machine Learning & IoT Integration

- Edge ML algorithms for sensory pattern detection.
- OpenCV computer vision for color & contour tracking.
- IoT architectures: MQTT pub/sub protocol.
- Wireless Wi-Fi telemetry streaming to cloud brokers.

WEEK 6

Data Processing & Robot Simulation

- Sensor noise filtering (moving average, Kalman basics).
- Multi-sensor data fusion algorithms.
- Physics-based robot simulation in Webots / Gazebo.
- Pre-deployment validation of navigation loops.

WEEK 7

Robotics Ethics, Safety & Security

- International safety standards (ISO 10218 / 13849).
- Hardware emergency stop & watchdog timers.
- Cyber-physical threat modeling (sensor spoofing, replay).
- Encrypted TLS communication & credential protection.

WEEK 8

Capstone Project & System Debugging

- Integration of autonomous mobile robot system.
- Obstacle avoidance, vision tracking & IoT telemetry.
- Systematic hardware/software troubleshooting.
- Final Assessment Test & Credential Certification.

Key Projects & Demonstrated Technical Mastery

- ✓ **Autonomous Navigation Engine:** Engineered reactive collision avoidance utilizing ultrasonic sensors mounted on pan-servos and bilateral IR detectors with 0% collision rate in dynamic obstacle simulation.
- ✓ **Edge Computer Vision & AI:** Integrated OpenCV edge vision pipeline processing 30 FPS camera feeds for color landmark segmentation and target tracking directly on embedded modules.
- ✓ **Real-Time Closed-Loop Kinematics:** Designed differential-drive kinematic algorithms with closed-loop PID RPM regulation, ensuring smooth acceleration profiles and wheel slippage mitigation.
- ✓ **Asynchronous IoT Telemetry Streaming:** Implemented an MQTT client transmitting JSON-encoded diagnostic packets (battery, motor RPM, sensor readings) over Wi-Fi to a cloud dashboard at 250ms intervals.
- ✓ **Hardened Cyber-Physical Security:** Embedded TLS encryption, replay-protected command tokens, watchdog fail-safes, and automatic motor cut-off upon communication loss (>1000ms).

Certification & Official Verification

OFFICIAL INSTITUTIONAL CREDENTIALS

- Internship Track: Advanced Robotics Using AI And IoT
- Curriculum Designation: ROBOTICS FUNDAMENTALS - AI AND IOT INTEGRATED
- Host Organization: EduSkills Academy (ISO 9001:2015 / ISO 20000-1:2018)
- Patronage: AICTE National Internship Portal (Ministry of Education)
- Offer Letter Ref No.: C16Y2026M091396204
- AICTE Student ID: STU69e08d9e3f2e51776323998
- Certificate ID: 2026-DFEBBA08EF
- Certificate Issue Date: September 10, 2026
- Authorizing Signatory: Mitu Swain (General Manager - L&D, EduSkills)
- Status: Verified & Successfully Completed

ACCREDITATION SUMMARY

- ★ National Recognition: Government-patronized AICTE internship credential validating practical engineering competencies.
- ★ Comprehensive Curriculum: Spans low-level electronics, embedded systems, kinematics, edge ML, and cloud IoT.
- ★ Rigorous Assessment: Evaluated through 8 continuous weekly milestone tests, lab submissions, and a comprehensive final examination.
- ★ Cyber-Physical Integration: Demonstrates multidisciplinary capability across robotics and cyber security disciplines.

Thank You!

The floor is open for technical questions, feedback, and discussion.

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